

SOLIDWORKS API (.NET, VBA, C#) — Reference (106/337): IFace2 (part 2 of 3)

SOLIDWORKS API Documentation (converted from CHM)

IFace2 (part 2 of 3) (Type)

GetTessTriStripSize Method (IFace2)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IFace2 Interface](#) : GetTessTriStripSize Method (IFace2)

Gets the array size required for [IFace2::GetTessTriStrips](#) and [IFace2::IGetTessTriStrips](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetTessTriStripSize() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Integer  
  
value = instance.GetTessTriStripSize()
```

C#

```
System.int GetTessTriStripSize()
```

C++/CLI

```
System.int GetTessTriStripSize();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array size

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTessTriStripSize](#).

Remarks

The return value from this method is the number of floats returned from [IFace2::GetTessTriStrips](#) and [IFace2::IGetTessTriStrips](#). That number is (1 + NumStrips + 3 * VertexCount).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTessNorms Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::IGetTessNorms Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStripNorms Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

GetTessTriStripTextures Method (IFace2)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IFace2 Interface](#) : GetTessTriStripTextures Method (IFace2)

Gets the texture coordinate components for each vertex on each triangle strip on this face.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetTessTriStripTextures() As System.Object
```

Visual Basic (Usage)

```
Dim instance As IFace2
Dim value As System.Object

value = instance.GetTessTriStripTextures()
```

C#

```
System.object GetTessTriStripTextures()
```

C++/CLI

```
System.Object^ GetTessTriStripTextures();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing the triangle strip texture coordinate components (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTessTriStripTextures](#).

Remarks

Before calling this method, call [IFace2::HasTextureCoordinates](#).

The format of the returned data is:

DWORD *NumStrips*

DWORD [] *VertexesPerStrip*

Double [] *TextureCoords*

where:

- NumStrips* = Number of tri strips on this face
- VertexPerStrip* = Array containing the number of vertex points on each tri strip
- TextureCoords* = Array from 0 to (*VertexPerStrip* - 1) containing the U,V texture coordinate components for each vertex on each tri strip

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IPartDoc::GetTessTriStripTextures Method \(\)](#)
[IFace2::GetTessNorms Method \(\)](#)
[IFace2::GetTessTextures Method \(\)](#)
[IFace2::GetTessTriangleCount Method \(\)](#)
[IFace2::GetTessTriangles Method \(\)](#)
[IFace2::GetTessTriStripEdges Method \(\)](#)
[IFace2::GetTessTriStripNorms Method \(\)](#)
[IFace2::GetTessTriStripSize Method \(\)](#)
[IFace2::GetTessTriStripSize Method \(\)](#)

Availability

SOLIDWORKS 2025 FCS, Revision Number 33

SOLIDWORKS API Help

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GetTexture Method (IFace2)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTexture Method (IFace2)

Gets the texture applied to this face in the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTexture( _  
    ByVal Config_name As System.String _  
) As Texture
```

Visual Basic (Usage)



```
Dim instance As IFace2
Dim Config_name As System.String
Dim value As Texture

value = instance.GetTexture(Config_name)
```

C#

```
Texture GetTexture(
    System.string Config\_name
)
```

C++/CLI

```
Texture^ GetTexture(
    & System.String^ Config\_name
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_name
Name of the configuration

Return Value

If texture is...	Then retval is...
Applied to this face in Config_name	Pointer to ITexture object
Not present	Null

 Visual Basic for Applications (VBA) Syntax

See [Face2::GetTexture](#).

 Example

[Change Texture on Face in Specified Configuration \(VBA\)](#)
[Change Texture on Face in Specified Configuration \(VB.NET\)](#)
[Change Texture on Face in Specified Configuration \(C#\)](#)

 See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::RemoveTexture Method](#)
[IFace2::SetTexture Method](#)
[IFace2::SetTextureByDisplayState Method](#)
[IFace2::RemoveTextureByDisplayState Method](#)
[IModelDocExtension::CreateTexture Method](#)
[IFace2::HasTextureCoordinates Method \(\)](#)

 Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

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GetTrackingIDs Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTrackingIDs Method (IFace2)

Gets the [tracking IDs assigned to this face](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function GetTrackingIDs( _
    ByVal TrackingCookie As System.Integer, _
    ByRef TrackingIDs As System.Object _
) As System.Integer
```

Visual Basic (Usage)



Copy Code

```
Dim instance As IFace2  
Dim TrackingCookie As System.Integer  
Dim TrackingIDs As System.Object  
Dim value As System.Integer  
  
value = instance.GetTrackingIDs(TrackingCookie, TrackingIDs)
```

```
C#  
  
System.int GetTrackingIDs(  
    System.int TrackingCookie,  
    out System.object TrackingIDs  
)
```

```
C++/CLI  
  
System.int GetTrackingIDs(  
& System.int TrackingCookie,  
& [Out] System.Object^ TrackingIDs  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie
Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)
TrackingIDs
Array of tracking IDs on this face

Return Value

Status as defined by [swTrackingIDError](#) e

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTrackingIDs](#).

Example

[Get, Set, and Remove Tracking IDs on Face \(VBA\)](#)

Remarks

See [Tracking IDs](#) for more information.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTrackingIDsCount Method](#)
[IFace2::IGetTrackingIDs Method](#)
[IFace2::RemoveTrackingID Method](#)
[IFace2::SetTrackingID Method](#)
[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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GetTrackingIDsCount Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTrackingIDsCount Method (IFace2)

Gets the number of tracking IDs on this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTrackingIDsCount( _  
    ByVal TrackingCookie As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)

 Copy Code

```
Dim instance As IFace2  
Dim TrackingCookie As System.Integer  
Dim value As System.Integer  
  
value = instance.GetTrackingIDsCount(TrackingCookie)
```

C#

```
System.int GetTrackingIDsCount(  
    System.int TrackingCookie  
)
```

C++/CLI

```
System.int GetTrackingIDsCount(  
& System.int TrackingCookie  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie
Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

Return Value

Number of tracking IDs on this face

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTrackingIDsCount](#).

Example

[Get, Set, and Remove Tracking IDs on Face \(VBA\)](#)

Remarks

Call this method before calling [IFace2::IGetTrackingIDs](#) to determine the size of the array for that method.
See [Tracking IDs](#) for more information.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetTrackingIDs Method](#)
[IFace2::RemoveTrackingID Method](#)
[IFace2::SetTrackingID Method](#)
[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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GetTrimCurves Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTrimCurves Method (IFace2)

Obsolete. Superseded by [IFace2::GetTrimCurves2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetTrimCurves( _  
    ByVal WantCubic As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim WantCubic As System.Boolean  
Dim value As System.Object  
  
value = instance.GetTrimCurves(WantCubic)
```

C#

```
System.object GetTrimCurves(  
    System.bool WantCubic  
)
```

C++/CLI

```
System.Object^ GetTrimCurves(  
& System.bool WantCubic  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubic

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTrimCurves](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)

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GetTrimCurves2 Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTrimCurves2 Method (IFace2)

Gets the definition of all of the entities that describe a trimmed face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTrimCurves2( _  
    ByVal WantCubic As System.Boolean, _  
    ByVal WantNRational As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim WantCubic As System.Boolean  
Dim WantNRational As System.Boolean  
Dim value As System.Object  
  
value = instance.GetTrimCurves2(WantCubic, WantNRational)
```

C#

```
System.object GetTrimCurves2(  
    System.bool WantCubic,  
    System.bool WantNRational  
)
```

C++/CLI

```
System.Object^ GetTrimCurves2(  
& System.bool WantCubic,  
& System.bool WantNRational  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubic

True if the trim curves are to be cubic, false if not

WantNRRational

True if the trim curves are to be non-rational, false if not

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTrimCurves2](#).

Remarks

This method returns an array containing the definition of all of the entities that describe a trimmed face. The data is arranged in collections of surface curves (curves defined in the 2D parametric space of the surface, or SP-curves) organized into loops (closed composite curves).

All SP-curves are returned as b-splines. To control the tolerance of the SP-curves, use [IModeler::GetToleranceValue](#) or [IModeler::SetToleranceValue](#).

A call to [IFace2::IGetTrimCurveSize2](#) must precede the call to [IFace2::GetTrimCurves2](#) to get the size of the array required and pre-process the trim curve data.

The surface and trim curve parameterization returned by this method might be different from the parameterization returned by other methods, such as [IFace2::GetUVBounds](#), [ISurface::Parameterization](#), [IEdge::GetCurveParams2](#), and so on.

If you set *WantCubic* to true, then this method returns the underlying Bsurface definition. If you want to use the Bsurface in combination with its trim curves, set *WantCubic* = True because it provides better alignment of the trim curves with the Bsurface as they are generated at the same time.

Each face typically has at least one outer loop and zero or more inner loops. Outer loops define the periphery of the face; inner loops define holes in the face. Outer loops are returned before inner loops when possible. If you need to be certain, use [ILoop2::IsOuter](#).

Periodic Faces

Periodic faces present many obstacles in translation. As a general rule, it is best to chop up all periodic surfaces by making a call to [IBody2::GetProcessedBody2](#). This method returns a new body to the caller with periodic faces chopped up into multiple faces. Processing this new body, instead of the original body, avoids many of the pitfalls associated with periodics.

If this face is periodic and crosses the "seam" (0 & 360 location), then this method splits the face into more than one trimmed surface for the purposes of returning this set of data. The first time that this method is called on any face, it returns data for the first trimmed surface on the face. Each subsequent call to this function on the same face returns data for the next trimmed surface, if one exists. The *packedDouble8* parameter determines the number of surfaces for this face.

To avoid the return of multiple surfaces per face, process the body returned from [IBody2::GetProcessedBody2](#).

Using the *Body2* object returned from [IBody2::GetProcessedBody2](#) usually prevents [IFace2::GetTrimCurves2](#) from returning multiple outer loops when it encounters certain periodic surfaces. In other words, it can prevent [IFace2::GetTrimCurves2](#) from generating multiple faces (each with an outer loop) for a particular periodic surface.

Edges

If your program uses calls to both [IFace2::GetEdges](#) and [IFace2::GetTrimCurves2](#) or to both [IFace2::EnumEdges](#) and [Face2::GetTrimCurves2](#), then:

[IFace2::GetTrimCurves2](#) stores a temporary buffer containing the edges for this face. If this temporary buffer exists, then calls to [IFace2::GetEdges](#) or [IFace2::EnumEdges](#) access the buffer. This is beneficial if you are attempting to align the trim curves of the face with their corresponding edges.

If your call to [IFace2::GetEdges](#) or [IFace2::EnumEdges](#) is made immediately following your call to [IFace2::GetTrimCurves2](#), then the order of the trim curves aligns with the order of the edges. However, if you are processing all the trim curves and then processing all the edges, your first call to access the edges uses the temporary buffer of edges generated in your last call to [IFace2::GetTrimCurves2](#).

Because your last call to [IFace2::GetTrimCurves2](#) and your current call to get the edges were probably made from different faces, you will see a mismatch of data. This mismatch of data might be seen when [IFace2::GetEdgeCount](#) does not match the array size from [IFace2::GetEdges](#) or it might be seen when the edge data does not match the face being examined. Therefore, calls to [IFace2::GetEdges](#) or [IFace2::EnumEdges](#) should be made in the same loop and immediately following your call to [IFace2::GetEdges](#). The temporary buffer of edges is destroyed after each call to [IFace2::GetEdges](#) or [IFace2::EnumEdges](#).

A corresponding edge for each trim curve might not exist. This is true for periodic surfaces for which no edge exists along the seam (0 and 360), yet this method returns a trim curve. You can detect this situation by comparing the number of edges on the face, [IFace2::GetEdgeCount](#), with the number of trim (SP) curves on the face, *PackedDouble1*.

To force edge generation along the seam of a periodic surface, use [IBody2::GetProcessedBody](#). This method creates a copy of the body with faces and surfaces chopped up. However, there are still certain situations (poles of a surface, cusps on a surface, and so on) in which this method returns a trim curve where no edge exists. In this situation, the list of edges returned from [IFace2::GetEdges](#) or [IFace2::EnumEdges](#) does not align with the list of trim curves. Again, you can detect this by comparing the value returned by [IFace2::GetEdgeCount](#) with *PackedDouble1*, but you must handle matching the edges with their corresponding trim curve.

Return Values

The data is returned in an array of doubles. Integer data is packed among the doubles in pairs.

The format of *faceData[]* is as follows:

[*packedDouble1*, *packedDouble2*[], *packedDouble3*[], *knotValues*[], *ctrlPointCoords*[], *packedDouble8*]

packedDouble1

An integer pair containing number of loops and total number of SP-curves (trim curves). The length of any edge list generated immediately after a call to [IFace2::GetTrimCurves2](#) will be equal to the number of SP-curves.

<i>packedDouble2[]</i>	Series of integer pairs containing the number of SP-curves in each loop. The first integer in each pair represents the number of curves in the odd loops; the second represents the even. The total number of integer pairs is half the number of loops, rounded up
<i>packedDouble3[]</i>	For each SP-curve, a set of two integer pairs. The first contains the order of the curve and a Boolean indicating if it is periodic. If the curve is periodic, it is clamped (that is, knots of multiplicity = order exists at each end of the curve). The second contains the dimension of the curve and the number of control points in it. If the dimension is 2, then the curve is non-rational; if the dimension is 3, then the curve is rational. NOTE: An SP-curve might be treated as periodic if it is closed and G1 continuous even if the b-spline representation returned is not periodic. For example, the periodic flag returned for the boundary curve of a sheet body with a circle as the boundary loop is true, because the circle is closed and is tangent-continuous at the start/end point, but the data returned for the b-spline representation is in the clamped format, i.e., for a non-periodic b-spline.
<i>knotValues[]</i>	Array of doubles containing the knot values for all the SP-curves in order. The number of knots for each curve is equal to the order of the curve plus the number of control points.
<i>ctrlPointCoords[]</i>	Array of doubles containing the coordinates of the control points for all the SP-curves in order. These are given as a sequence of (U,V) pairs if non-rational, and (U,V,W) triples if rational, where W is the homogeneous coordinate known as the weight.
<i>packedDouble8</i>	Integer pair as follows: the first integer is the total number of trimmed surfaces being output for the given face (the modeler sometimes outputs more than one trimmed surface for a face). The second integer indicates the index of the current trimmed surface being output. All the data in the <i>faceData[]</i> array is for the current trimmed surface. Therefore, if <i>packedDouble8</i> says there are two trim surfaces, you should use the same IFace2 pointer and call IFace2::GetTrimCurves2 again to get the data for the second trim surface.

If b-splines were requested (WantCubic = True), then the underlying Bsurface is also returned. The *faceData[]* array would be as follows:

[*packedDouble1*, *packedDouble2[]*, *packedDouble3[]*, *knotValues[]*, *ctrlPointCoords[]*, *packedDouble4*, *packedDouble5*, *packedDouble6*, *packedDouble7*, *surfaceKnotValuesU[]*, *surfaceKnotValuesV[]*, *surfaceCtrlPointCoords[]*, *packedDouble8*]

<i>packedDouble1</i>	See previous definition.
<i>packedDouble2[]</i>	See previous definition.
<i>packedDouble3[]</i>	See previous definition.
<i>knotValues[]</i>	See previous definition.
<i>ctrlPointCoords[]</i>	See previous definition.
<i>packedDouble4</i>	Integer pair containing the dimension of the surface (3 is non-rational and 4 is rational). The second member of the pair is not used and contains a 0.
<i>PackedDouble5</i>	Integer pair containing the U-order and the V-order of the surface.
<i>PackedDouble6</i>	Integer pair containing the number of control points in U and the number of control points in V.
<i>packedDouble7</i>	Integer pair containing Booleans indicating if the surface is periodic in U and in V. If the surface is periodic, it is clamped (that is, knots of multiplicity = order exist at each end of the surface).
<i>surfaceKnotValuesU[]</i>	Array of doubles containing the knot values in U for the surface. The number of knots in U is equal to the order in U plus the number of control points in U.
<i>surfaceKnotValuesV[]</i>	Array of doubles containing the knot values in V as above for U.
<i>surfaceCtrlPointCoords[]</i>	Array of doubles containing the coordinates of the control points for the surface. These are given as a sequence of (X,Y,Z) triples if non-rational, and (X,Y,Z,W) tuples if rational, where W is the homogeneous coordinate known as the weight. The coordinates are given in row-wise order (U varying most quickly).
<i>packedDouble8</i>	See previous definition.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTrimCurveTopology Method](#)
[IFace2::GetTrimCurveTopologyCount Method](#)
[IFace2::GetTrimCurveTopologyTypes Method](#)
[IFace2::IGetTrimCurveSize2 Method](#)
[IFace2::IGetTrimCurveTopology Method](#)
[IFace2::IGetTrimCurveTopologyTypes Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0



SolidWorks.Interop.sldworks Namespace > [IFace2 Interface](#) : GetTrimCurveTopology Method (IFace2)

Gets the trim curve topology for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTrimCurveTopology() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetTrimCurveTopology()
```

C#

```
System.object GetTrimCurveTopology()
```

C++/CLI

```
System.Object^ GetTrimCurveTopology();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of coedges, vertices, and NULLs (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTrimCurveTopology](#).

Remarks

This method returns a list of coedges, vertices, and NULLs. You must call [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) before calling this method.

This method returns a list of proper edges corresponding to the list of SP curves from [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). Otherwise, this method prompts the user for the list of edges that the software thinks are on the face. These two lists differ in order and content.

Use [IFace2::GetTrimCurveTopologyTypes](#) or [IFace2::IGetTrimCurveTopologyTypes](#) to get an array of types corresponding to the returned objects.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IFace2::IGetTrimCurveTopology Method](#)
[IFace2::GetTrimCurveTopologyCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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GetTrimCurveTopologyCount Method (IFace2)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IFace2 Interface](#) : GetTrimCurveTopologyCount Method (IFace2)

Gets the number of elements in the trim curve topology for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTrimCurveTopologyCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2
Dim value As System.Integer

value = instance.GetTrimCurveTopologyCount()
```

C#

```
System.int GetTrimCurveTopologyCount()
```

C++/CLI

```
System.int GetTrimCurveTopologyCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of elements

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTrimCurveTopologyCount](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTrimCurves2 Method](#)
[IFace2::GetTrimCurveTopology Method](#)
[IFace2::GetTrimCurveTopologyTypes Method](#)
[IFace2::IGetTrimCurveSize2 Method](#)
[IFace2::IGetTrimCurveTopology Method](#)
[IFace2::IGetTrimCurveTopologyTypes Method](#)

Availability

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GetTrimCurveTopologyTypes Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTrimCurveTopologyTypes Method (IFace2)

Gets the types of elements in the trim curve topology for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTrimCurveTopologyTypes() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2
Dim value As System.Object

value = instance.GetTrimCurveTopologyTypes()
```

C#

```
System.object GetTrimCurveTopologyTypes()
```

C++/CLI

```
System.Object^ GetTrimCurveTopologyTypes();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of longs or integers (see [Long vs. Integer](#)) indicating topology types as defined in [swSelectType_e](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTrimCurveTopologyTypes](#).

Remarks

Use [IFace2::GetTrimCurveTopology](#) or [IFace2::IGetTrimCurveTopology](#) to get the trim curve topology for this face.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTrimCurves2 Method](#)
[IFace2::GetTrimCurveTopologyCount Method](#)
[IFace2::IGetTrimCurveSize2 Method](#)
[IFace2::IGetTrimCurveTopologyTypes Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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GetUVBounds Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetUVBounds Method (IFace2)

Gets the values that describe the U, V bounds of this face.

NET Syntax

Visual Basic (Declaration)

```
Function GetUVBounds() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetUVBounds()
```

C#

```
System.object GetUVBounds()
```

C++/CLI

```
System.Object^ GetUVBounds();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetUVBounds](#).

Example

[Create Trimmed Curve \(VBA\)](#)
[Create Trimmed Curve \(VB.NET\)](#)
[Create Trimmed Curve \(C#\)](#)

Remarks

The return values bound an area of the surface in which the face is defined. You can determine the surface U, V bounds using [ISurface::Parameterization](#) or [ISurface::IParameterization](#).

The returned data is an array of 4 doubles arranged in the following order:

- retval[0] - Minimum U parameter of this face
- retval[1] - Maximum U parameter of this face
- retval[2] - Minimum V parameter of this face
- retval[3] - Maximum V parameter of this face

The minimum parameters are always less than the maximum parameters, and the range (for example, retval[1] - retval[0] and retval[3]-retval[2]) is always less than or equal to the U and V range of the underlying surface.

For surfaces with periodic parameters, the face parameter box can never be larger than the period of the corresponding surface parameter.

```
uRange[0] <= retval[0] < uRange[1]
```

```
vRange[0] <= retval[2] < vRange[1]
```

where uRange and vRange describe the UV range of the surface

Therefore, a face that straddles the boundary of a periodic parameter has an upper parameter value for the face that is greater than the upper parameter range of the surface.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetUVBounds Method](#)
[IBody2::GetProcessedBody2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

HasMaterialPropertyValues Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : HasMaterialPropertyValues Method (IFace2)

Gets whether this face has an appearance.

.NET Syntax

Visual Basic (Declaration)

```
Function HasMaterialPropertyValues() As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IFace2  
Dim value As System.Boolean  
  
value = instance.HasMaterialPropertyValues()
```

C#

```
System.bool HasMaterialPropertyValues()
```

C++/CLI

```
System.bool HasMaterialPropertyValues();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this face has an appearance, false if not

Visual Basic for Applications (VBA) Syntax

See [Face2::HasMaterialPropertyValues](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::MaterialPropertyValues Property \(\)](#)
[IFace2::GetMaterialPropertyValues2 Method \(\)](#)
[IFace2::RemoveMaterialProperty2 Method \(\)](#)
[IFace2::SetMaterialPropertyValues2 Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

HasTextureCoordinates Method (IFace2)

[See Also](#)



 .NET Syntax

Visual Basic (Declaration)

```
Function HasTextureCoordinates() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Boolean  
  
value = instance.HasTextureCoordinates()
```

C#

```
System.bool HasTextureCoordinates()
```

C++/CLI

```
System.bool HasTextureCoordinates();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if texture coordinates exist, false if not

 Visual Basic for Applications (VBA) Syntax

See [Face2::HasTextureCoordinates](#).

 Remarks

Call this method before calling [IFace2::GetTessTriStripTextures](#).
Texture coordinates are generated only when mapping type is set to surface.

 See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTessTextures Method \(\)](#)
[IFace2::GetTexture Method \(\)](#)
[IFace2::RemoveTexture2 Method \(\)](#)
[IFace2::RemoveTextureByDisplayState Method \(\)](#)
[IFace2::SetTexture Method \(\)](#)
[IFace2::SetTextureByDisplayState Method \(\)](#)
[IFace2::GetSurface Method \(\)](#)

 Availability

SOLIDWORKS 2025 SP01, Revision Number 23.0



 .NET Syntax

Visual Basic (Declaration)

```
Sub Highlight(  
    ByVal State As System.Boolean  
)
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim State As System.Boolean
```

```
instance.Highlight(State)
```

C#

```
void Highlight(  
    System.bool State  
)
```

C++/CLI

```
void Highlight(  
&    System.bool State  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

State

True adds highlighting, false removes highlighting

Visual Basic for Applications (VBA) Syntax

See [Face2::Highlight](#).

Example

[Roll Back Model \(C#\)](#)
[Roll Back Model \(VB.NET\)](#)
[Roll Back Model \(VBA\)](#)

Remarks

Highlighting remains in effect until the model is rebuilt or redrawn.

This method is not supported for faces obtained from reference surface bodies.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IHighlight Method](#)
[IEdge::IHighlight Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateSheetBody Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : ICreateSheetBody Method (IFace2)

Creates a sheet body from this face.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSheetBody() As Body2
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As Body2  
  
value = instance.ICreateSheetBody()
```

C#

```
Body2 ICreateSheetBody()
```

C++/CLI

```
Body2^ ICreateSheetBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

New sheet [body](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::ICreateSheetBody](#).

Remarks

A sheet body is a surface body.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::CreateSheetBody Method](#)
[IFace2::CreateSheetBodyByFaceExtension Method](#)
[IFace2::ICreateSheetBodyByFaceExtension Method](#)
[IBody2::DeleteFacesMakeSheetBodies Method](#)
[IBody2::IDeleteFacesMakeSheetBodies Method](#)
[IBody2::IDeleteFacesMakeSheetBodiesCount Method](#)
[IModeler::CreateBodiesFromSheets2 Method](#)
[IModeler::ICreateBodiesFromSheets2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateSheetBodyByFaceExtension Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : ICreateSheetBodyByFaceExtension Method (IFace2)

Creates a sheet body by extending the face.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSheetBodyByFaceExtension( _  
    ByRef BoxLowIn As System.Double, _  
    ByRef BoxHighIn As System.Double _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim BoxLowIn As System.Double  
Dim BoxHighIn As System.Double  
Dim value As Body2
```

```
value = instance.ICreateSheetBodyByFaceExtension(BoxLowIn, BoxHighIn)
```

C#

```
Body2 ICreateSheetBodyByFaceExtension(  
    ref System.double BoxLowIn,  
    ref System.double BoxHighIn  
)
```

C++/CLI

```
Body2^ ICreateSheetBodyByFaceExtension(  
& System.double% BoxLowIn,  
& System.double% BoxHighIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BoxLowIn

Pointer to an array of 3 doubles (x,y,z)

BoxHighIn

Pointer to an array of 3 doubles (x,y,z)

Return Value

Pointer to the newly created sheet [body](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::ICreateSheetBodyByFaceExtension](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IBody2::DeleteFacesMakeSheetBodies Method](#)
[IBody2::IDeleteFacesMakeSheetBodies Method](#)
[IBody2::IDeleteFacesMakeSheetBodiesCount Method](#)
[IFace2::CreateSheetBody Method](#)
[IFace2::CreateSheetBodyByFaceExtension Method](#)
[IFace2::ICreateSheetBody Method](#)
[IModeler::CreateBodiesFromSheets2 Method](#)
[IModeler::ICreateBodiesFromSheets2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetAllDecalProperties Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetAllDecalProperties Method (IFace2)

Gets the decal properties applied to this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetAllDecalProperties( _  
    ByVal Count As System.Integer _  
) As FaceDecalProperties
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim Count As System.Integer  
Dim value As FaceDecalProperties  
  
value = instance.IGetAllDecalProperties(Count)
```

C#

```
FaceDecalProperties IGetAllDecalProperties(  
    System.int Count  
)
```

C++/CLI

```
FaceDecalProperties^ IGetAllDecalProperties(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of decals applied to this face

Return Value

- in-process, unmanaged C++: Pointer to an array of [decal properties applied to this face](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IFace2::GetDecalsCount](#) to get the value of Count.

 See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetAllDecalProperties Method](#)
[IFace2::IGetDecalProperties Method](#)

 Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetBody Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetBody Method (IFace2)

Gets the body containing this face.

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetBody() As Body2
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As Body2  
  
value = instance.IGetBody()
```

C#

```
Body2 IGetBody()
```

C++/CLI

```
Body2^ IGetBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [body](#)

 Visual Basic for Applications (VBA) Syntax

See [Face2::IGetBody](#).

 Example

[Enumerate Bodies \(C++\)](#)

 See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetBody Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetBox Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetBox Method (IFace2)

Gets the box boundaries for this face.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetBox() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Double  
  
value = instance.IGetBox()
```

C#

```
System.double IGetBox()
```

C++/CLI

```
System.double IGetBox();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Two diagonal points that bound the component in the form of a pointer to an array of 6 doubles

Visual Basic for Applications (VBA) Syntax

See [Face2::IGetBox](#).

Remarks

IMPORTANT: The values returned are approximate and should not be used for comparison or calculation purposes. Furthermore, the bounding box may vary after rebuilding the model.

The resulting box encloses the object, but it might not be the tightest box.

This method returns X,Y,Z points that are the lower and upper diagonal corners that bound the face with the box sides parallel to the X, Y and Z axes. The box dimensions enclose the face and are typically close to the minimum possible size, but this is not guaranteed.

The return value is an array of doubles as follows:

[XCorner1, YCorner1, ZCorner1, XCorner2, YCorner2, ZCorner2]

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetBox Method](#)
[IAssemblyDoc::GetBox Method](#)
[IAssemblyDoc::IGetBox Method](#)
[IBody2::GetBodyBox Method](#)
[IBody2::IGetBodyBox Method](#)
[IComponent2::GetBox Method](#)
[IComponent2::IGetBox Method](#)
[IFeature::GetBox Method](#)
[IFeature::IGetBox Method](#)
[IModelDocExtension::SetVisibleBox Method](#)
[IModelDocExtension::RemoveVisibleBox Method](#)
[IModelDocExtension::GetVisibleBox Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetClosestPointOn Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetClosestPointOn Method (IFace2)

Uses the X,Y,Z input point to determine the closest point on the face.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetClosestPointOn( _  
    ByVal X As System.Double, _
```

```
ByVal Y As System.Double, _  
ByVal Z As System.Double _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Double  
  
value = instance.IGetClosestPointOn(X, Y, Z)
```

C#

```
System.double IGetClosestPointOn(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.double IGetClosestPointOn(  
&    System.double X,  
&    System.double Y,  
&    System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X value of the input point

Y
Y value of the input point

Z
Z value of the input point

Return Value

- in-process, unmanaged C++: Pointer to an array of five doubles representing the X,Y,Z point on the face followed by the U, V parameter on the face that is closest to the input point
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This method returns only one point, regardless of how many points achieve the minimum distance.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetClosestPointOn Method](#)
[IFace2::IGetProjectedPointOn Method](#)
[ICurve::IGetClosestPointOn Method](#)
[ICurve::IGetClosestPointOn Method](#)
[IEdge::IGetClosestPointOn Method](#)
[IEdge::IGetClosestPointOn Method](#)
[ISurface::IGetClosestPointOn Method](#)
[ISurface::IGetClosestPointOn Method](#)
[IVertex::IGetClosestPointOn Method](#)
[IVertex::IGetClosestPointOn Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDecalProperties Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetDecalProperties Method (IFace2)

Gets the properties of the specified decal on this face.

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetDecalProperties( _  
    ByVal PDecal As Decal, _  
    ByRef pFaceDecalProperties As FaceDecalProperties _  
)
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim PDecal As Decal  
Dim pFaceDecalProperties As FaceDecalProperties  
  
instance.IGetDecalProperties(PDecal, pFaceDecalProperties)
```

C#

```
void IGetDecalProperties(  
    Decal PDecal,  
    ref FaceDecalProperties pFaceDecalProperties  
)
```

C++/CLI

```
void IGetDecalProperties(  
    & Decal^ PDecal,  
    & FaceDecalProperties^% pFaceDecalProperties  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PDecal

[Decal](#) whose properties to get

pFaceDecalProperties

- in-process, unmanaged C++: Pointer to an array of [decal properties](#)

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

See Also

[IFace2 Interface](#)

[IFace2 Members](#)

[IFace2::GetAllDecalProperties Method](#)

[IFace2::GetDecalsCount Method](#)

[IFace2::IGetAllDecalProperties Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetEdges Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetEdges Method (IFace2)

Get the edges bounding this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEdges() As Edge
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As Edge  
  
value = instance.IGetEdges()
```

C#

```
Edge IGetEdges()
```

C++/CLI

```
Edge^ IGetEdges() ;
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of [edges](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

If you want to align the edges with their corresponding trim curves, then call this method immediately following a call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). This causes the order of the trim curves to align with the order of the edges. For Dispatch implementations, use [IFace2::GetEdges](#)

To elaborate, any call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) stores a temporary buffer containing the edges for this face. If this temporary buffer exists, then [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#) access the buffer. This is useful when you want to align the trim curves of the face with their corresponding edges. If [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#) is called immediately after [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#), then the order of the trim curves align with the order of the edges. However, if you are processing all the trim curves and then processing all the edges, your first call to [IFace2::EnumEdges](#) uses the temporary buffer of edges generated by your last call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). Because your last call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) and your current call to [IFace2::EnumEdges](#) are probably made from different faces, the data does not match. You can see this mismatch of data when [IFace2::GetEdgeCount](#) does not match the number of enumerated edges, or when the edge data does not match the examined face. Therefore, call [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#) in the same loop and immediately after your call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). The temporary buffer of edges is destroyed after each call to [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFaces2::EnumEdges](#).

If you are using [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#), the number of trim curves on a face might vary from the number of edges on the face. If you call [IFace2::EnumEdges](#) immediately after calling [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#), then you are getting a list of edges generated by [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). Because this list of generated edges might vary from the list of edges that are typically returned from this face, do not use [IFace2::GetEdgeCount](#) to determine the length of the enumerated edge list. Instead, use one of the return values from [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) to determine the size of the enumerated edge list. [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) returns the number of trim (SP) curves in the first array position. This is the value you can use to loop through your enumerated list of edges.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFacetData Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetFacetData Method (IFace2)

Obsolete. Superseded by [IFace2::GetTessNormals](#), [IFace2::IGetTessNormals](#), [IFace2::GetTessTextures](#), [IFace2::IGetTessTextures](#), [IFace2::GetTessTriangleCount](#), [IFace2::GetTessTriangles](#), [IFace2::IGetTessTriangles](#), [IFace2::GetTessTriStripEdges](#), [IFace2::IGetTessTriStripEdges](#), [IFace2::GetTessTriStripNormals](#), [IFace2::IGetTessTriStripNormals](#), [IFace2::GetTessTriStrips](#), and [IFace2::IGetTessTriStrips](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetFacetData( _  
    ByVal FacetMesh As System.Integer, _  
    ByRef NFacets As System.Integer, _  
    ByRef NStrips As System.Integer, _  
    ByRef StripVertexNums As System.Integer, _  
    ByRef VertexCoords As System.Single, _  
    ByRef NormalCoords As System.Single _  
)
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim FacetMesh As System.Integer  
Dim NFacets As System.Integer  
Dim NStrips As System.Integer  
Dim StripVertexNums As System.Integer  
Dim VertexCoords As System.Single  
Dim NormalCoords As System.Single  
  
instance.IGetFacetData(FacetMesh, NFacets, NStrips, StripVertexNums, VertexCoords, NormalCoords)
```

C#

```
void IGetFacetData(
    System.int FacetMesh,
    ref System.int NFacets,
    ref System.int NStrips,
    ref System.int StripVertexNums,
    ref System.float VertexCoords,
    ref System.float NormalCoords
)
```

C++/CLI

```
void IGetFacetData(
    & System.int FacetMesh,
    & System.int% NFacets,
    & System.int% NStrips,
    & System.int% StripVertexNums,
    & System.float% VertexCoords,
    & System.float% NormalCoords
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

- FacetMesh*
- NFacets*
- NStrips*
- StripVertexNums*
- VertexCoords*
- NormalCoords*

 Visual Basic for Applications (VBA) Syntax

See [Face2::IGetFacetData](#).

 See Also

- [IFace2 Interface](#)
- [IFace2 Members](#)

IGetFeature Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetFeature Method (IFace2)

Gets the feature that owns this face.

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetFeature() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFace2
Dim value As Feature

value = instance.IGetFeature()
```

C#

```
Feature IGetFeature()
```

C++/CLI

```
Feature^ IGetFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the owning [feature](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::IGetFeature](#).

Remarks

In SOLIDWORKS, a face:

- is the result of evaluating a feature.
- can be owned by several features.

[IFeature::GetFaces](#) or [IFeature::IGetFaces2](#) returns all of the faces owned by a feature. This is different from the faces highlighted in the user interface when the feature is selected. The user interface filters out multiple feature faces. This filter is only for display purposes.

To filter out multiple feature faces using the SOLIDWORKS API, you must call [IFace2::GetFeature](#) or [IFace2::IGetFeature](#). Only the oldest feature from the face is returned, that is, the first owning feature in the FeatureManager design tree.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetFeatureId Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstLoop Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetFirstLoop Method (IFace2)

Gets the first loop in this face, which is not necessarily the outer loop.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstLoop() As Loop2
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As Loop2  
  
value = instance.IGetFirstLoop()
```

C#

```
Loop2 IGetFirstLoop()
```

C++/CLI

```
Loop2^ IGetFirstLoop();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the first [loop](#).

Visual Basic for Applications (VBA) Syntax

See [Face2::IGetFirstLoop](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetFirstLoop Method](#)
[IFace2::IGetLoopCount Method](#)
[ILoop2::GetNext Method](#)
[ILoop2::IGetNext Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetLoops Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetLoops Method (IFace2)

Gets all of the loops on this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLoops( _  
    ByVal NumberOfLoops As System.Integer _  
) As Loop2
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim NumberOfLoops As System.Integer  
Dim value As Loop2  
  
value = instance.IGetLoops(NumberOfLoops)
```

C#

```
Loop2 IGetLoops(  
    System.int NumberOfLoops  
)
```

C++/CLI

```
Loop2^ IGetLoops(  
    & System.int NumberOfLoops  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumberOfLoops
Number of loops on the face

Return Value

- in-process, unmanaged C++: Pointer to an array of all of the [loops](#) on the face (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IFace2::GetLoopCount](#) to get NumberOfLoops.

If a face has one or more holes on it, then this method gets a loop for the face and a loop for every hole on the face.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetLoops Method](#)
[IFace2::GetFirstLoop Method](#)
[IFace2::IGetFirstLoop Method](#)
[Loop2::GetNext Method](#)
[Loop2::IGetNext Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMaterialPropertyValues2 Method (IFace2)

[See Also](#)



Gets the material property values for this face.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetMaterialPropertyValues2( _
    ByVal Config\_opt As System.Integer, _
    ByVal Config\_count As System.Integer, _
    ByRef Config\_names As System.String _
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IFace2
Dim Config_opt As System.Integer
Dim Config_count As System.Integer
Dim Config_names As System.String
Dim value As System.Double

value = instance.IGetMaterialPropertyValues2(Config_opt, Config_count, Config_names)
```

C#

```
System.double IGetMaterialPropertyValues2(
    System.int Config\_opt,
    System.int Config\_count,
    ref System.string Config\_names
)
```

C++/CLI

```
System.double IGetMaterialPropertyValues2(
    & System.int Config\_opt,
    & System.int Config\_count,
    & System.String^% Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt
Configuration option as defined in [swInConfigurationOptions_e](#)
Config_count
Number of configurations
Config_names
Array of configuration names (see **Remarks**)

Return Value

- in-process, unmanaged C++: Pointer to an array of 9 doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The material property values returned include the color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission.

If you set *Config_opt* to *swThisConfiguration* or *swAllConfiguration*, then pass *Nothing* because *Config_names* is ignored.

The format of *material_values* is an array of doubles:

[*R*, *G*, *B*, *Ambient*, *Diffuse*, *Specular*, *Shininess*, *Transparency*, *Emission*]

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetMaterialPropertyValues2 Method](#)
[IFace2::IRemoveMaterialProperty2 Method](#)
[IFace2::ISetMaterialPropertyValues2 Method](#)
[IFace2::RemoveMaterialProperty2 Method](#)
[IFace2::SetMaterialPropertyValues2 Method](#)
[IFace2::IMaterialPropertyValues Property](#)
[IFace2::MaterialPropertyValues Property](#)
[IFeature::GetMaterialPropertyValues2 Method](#)
[IFeature::IGetMaterialPropertyValues2 Method](#)
[IModelDoc2::IMaterialPropertyValues Property](#)
[IModelDoc2::MaterialPropertyValues Property](#)
[IModelDocExtension::IGetMaterialPropertyValues Method](#)
[IModelDocExtension::GetMaterialPropertyValues Method](#)

[IPartDoc::MaterialPropertyValues Property](#)
[IPartDoc::IMaterialPropertyValues Property](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

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IGetNextFace Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetNextFace Method (IFace2)

Gets the next face in a body.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetNextFace() As Face2
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As Face2
```

```
value = instance.IGetNextFace()
```

C#

```
Face2 IGetNextFace()
```

C++/CLI

```
Face2^ IGetNextFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the next [face](#) in a body

Visual Basic for Applications (VBA) Syntax

See [Face2::IGetNextFace](#).

Example

[Traverse Bodies \(C++\)](#)

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetNextFace Method](#)
[IBody2::GetFirstFace Method](#)
[IBody2::GetFirstSelectedFace Method](#)
[IBody2::IGetFirstFace Method](#)
[IBody2::IGetFirstSelectedFace Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPatternSeedFeature Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetPatternSeedFeature Method (IFace2)

Gets the seed feature of a pattern.

.NET Syntax

Visual Basic (Declaration)

Function IGetPatternSeedFeature() As Feature

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As Feature  
  
value = instance.IGetPatternSeedFeature()
```

C#

```
Feature IGetPatternSeedFeature()
```

C++/CLI

```
Feature^ IGetPatternSeedFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Seed feature of the current face; if the face does not belong to a pattern, this method returns NULL

Visual Basic for Applications (VBA) Syntax

See [Face2::IGetPatternSeedFeature](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetPatternSeedFeature Method](#)
[ModelDoc2::EditSeedFeat Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetSilhoutteEdgeCount Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetSilhoutteEdgeCount Method (IFace2)

Gets the number of silhouette edges for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSilhoutteEdgeCount( _  
    ByRef Root As System.Double, _  
    ByRef Normal As System.Double _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim Root As System.Double  
Dim Normal As System.Double  
Dim value As System.Integer  
  
value = instance.IGetSilhoutteEdgeCount(Root, Normal)
```

C#

```
System.int IGetSilhoutteEdgeCount(  
    ref System.double Root,  
    ref System.double Normal  
)
```

C++/CLI

```
System.int IGetSilhoutteEdgeCount(  
& System.double% Root,
```

```
& System.double% Normal
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Root

Array of doubles defining the root point

Normal

Array of doubles defining the direction vector

Return Value

Number of silhouette edges

Visual Basic for Applications (VBA) Syntax

See [Face2::IGetSilhoutteEdgeCount](#).

Remarks

This method calculates the number of edges returned by [IFace2::IGetSilhoutteEdges](#) for a given vector root point (root) and a vector direction (normal). Call this method before calling IFace2::IGetSilhoutte.

See Also

[IFace2 Interface](#)

[IFace2 Members](#)

[IFace2::GetSilhoutteEdgesVB Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetSilhoutteEdges Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetSilhoutteEdges Method (IFace2)

Gets the silhouette edges for this face with the specified root point and in the specified direction.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSilhoutteEdges( _
    ByRef Root As System.Double, _
    ByRef Normal As System.Double _
) As Edge
```

Visual Basic (Usage)



```
Dim instance As IFace2
Dim Root As System.Double
Dim Normal As System.Double
Dim value As Edge

value = instance.IGetSilhoutteEdges(Root, Normal)
```

C#

```
Edge IGetSilhoutteEdges(
    ref System.double Root,
    ref System.double Normal
)
```

C++/CLI

```
Edge^ IGetSilhoutteEdges(
    & System.double% Root,
    & System.double% Normal
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Root

Array of doubles defining the root point

Normal

Array of doubles defining the direction vector

Return Value

- in-process, unmanaged C++: Pointer to an array to hold the edge pointers
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

These edges are not added to the face and cannot be found using [IFace2::IGetEdges](#). They are created and handed back to the caller as an array of edges packed with edges.

The vector root point (root) and a vector direction (normal) define the orientation for the silhouette edge creation.

The return array has two elements for each edge: the first is the silhouette edge and the second is not used. To iterate through the edges, an application needs to step through every second element.

Before calling this method, call [IFace2::IGetSilhouetteEdgeCount](#) to get the size of array required to hold the edges.

The returned edges are transient and cannot be selected.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetSilhouetteEdgesVB Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetSurface Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetSurface Method (IFace2)

Gets the surface referenced by this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSurface() As Surface
```

Visual Basic (Usage)

```
Dim instance As IFace2  
Dim value As Surface  
  
value = instance.IGetSurface()
```

C#

```
Surface IGetSurface()
```

C++/CLI

```
Surface^ IGetSurface();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the underlying [surface](#) for this face

Visual Basic for Applications (VBA) Syntax

See [Face2::IGetSurface](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetSurface Method](#)
[IFace2::AttachSurface Method](#)
[IFace2::DetachSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetTessNorms Method (IFace2)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IFace2 Interface](#) : IGetTessNorms Method (IFace2)

Gets the normal vector for each of the triangles that make up the shaded picture tessellation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessNorms() As System.Single
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Single  
  
value = instance.IGetTessNorms()
```

C#

```
System.float IGetTessNorms()
```

C++/CLI

```
System.float IGetTessNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of floats (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The data returned by this method takes the following format:

- float x, y, z - First point's unit normal
- float x, y, z - Second point's unit normal
- float x, y, z - Third point's unit normal

where this data repeats itself for the set of triangles on this IFace2.

The total size of the data is [9 x sizeof(float) x (number of triangles)].

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTessNorms Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStrips Method](#)
[IFace2::GetTessTriStripSize Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStripNorms Method](#)
[IFace2::IGetTessTriStrips Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetTessTextures Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetTessTextures Method (IFace2)

Gets the texture coordinates for the triangles.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTextures() As System.Single
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Single  
  
value = instance.IGetTessTextures()
```

C#

```
System.float IGetTessTextures()
```

C++/CLI

```
System.float IGetTessTextures();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of texture coordinates for the triangles
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetTessNormals Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStripNormals Method](#)
[IFace2::IGetTessTriStrips Method](#)
[IFace2::GetTessNormals Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripNormals Method](#)
[IFace2::GetTessTriStrips Method](#)
[IFace2::GetTessTriStripSize Method](#)

Availability

SOLIDWORKS 2005 SP2, Revision Number 13.2

SOLIDWORKS API Help

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IGetTessTriangles Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetTessTriangles Method (IFace2)

Gets the triangles that make up the shaded picture tessellation for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriangles( _  
    ByVal NoConversion As System.Boolean _  
) As System.Single
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim NoConversion As System.Boolean  
Dim value As System.Single
```

```
value = instance.IGetTessTriangles(NoConversion)
```

C#

```
System.float IGetTessTriangles(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.float IGetTessTriangles(  
&    System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True prohibits conversion to user units from system units, false does not

Return Value

- in-process, unmanaged C++: Pointer to array of floats (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

These triangles are intended for graphics display purposes and do not represent a tessellation that can be used, for example, by a machining application. If you need the kind of accuracy associated with a machining product, traverse the body faces and extract the topology and geometry data to create your own faceting.

The format of the returned data is:

float x, y, z - First point in meters

float x, y, z - Second point in meters

float x, y, z - Third point in meters

where this data repeats itself for each of the triangles on this face.

The total size of the data is [9 x sizeof(float) x (number of triangles)].

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTessNormals Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripNormals Method](#)
[IFace2::GetTessTriStrips Method](#)
[IFace2::GetTessTriStripSize Method](#)
[IFace2::IGetTessNormals Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStripNormals Method](#)
[IFace2::IGetTessTriStrips Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetTessTriStripEdges Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetTessTriStripEdges Method (IFace2)

Gets the edge ID for each of the edges in the triangles that make up the tessellation for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriStripEdges() As System.Integer
```

Visual Basic (Usage)

Copy Code

```
Dim instance As IFace2
Dim value As System.Integer

value = instance.IGetTessTriStripEdges()
```

C#

```
System.int IGetTessTriStripEdges()
```

C++/CLI

```
System.int IGetTessTriStripEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of longs containing the list of edge IDs for this face (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Returns an array of long values that indicate which edges of the given triangle strip represent a face boundary. The array takes the form of:

```
[nStrips,
 stripEdgeCount1, stripEdgeCount2,..., stripVertCountN,
 strip1[triStripFlag, stripEdgeCount1 elements],
 strip2[triStripFlag, stripEdgeCount2 elements],...,
 stripN[triStripFlag, stripEdgeCountN elements&cd;
```

where:

nStrips = The number of triangle strips on the face.
stripEdgeCountN = The number of triangle edges for the *N*th triangle strip + 1.
stripN = A sub-array that consists of the *triStripFlag* and an array of *stripEdgeCountN* *edgeIds*.
triStripFlag = Indicates if the triangle strip is a strip (=1) or if the triangle strip is a triangle (=0).
edgeId = An identifier with an arbitrary value that represents the edge of the face. Each edge has a unique *edgeId*.

```
2---4---6---8---10
 / | / | / | / |
 / | / | / | / |
 / | / | / | / |
 / | / | / | / |
 / | / | / | / |
1---3---5---7---9
```

If you have this triangle strip, then you get a *stripVertexCount* of 10 using [IFace2::GetTessTriStrips](#) or [IFace2::IGetTessTriStrips](#). [IFace2::GetTessTriStripEdges](#) and [IFace2::IGetTessTriStripEdges](#) get a *stripEdgeCount* of 18. This means that for each triangle strip *stripEdgeCount* - 1 = 2(*stripVertexCount* - 2) + 1. The first element represents a flag indicating if the current strip is a triangle strip or a simple triangle, 1 or 0, respectively.

If one of the triangle strip edges lies on a face edge, then the value of *stripN*[*i*] = *edgeId*. If the triangle strip edge does not lie on a face edge, then the value of *stripN*[*i*] = 0.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTessNorms Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStripSize Method](#)
[IFace2::IGetTessNorms Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStripNorms Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetTessTriStripEdgeSize Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : [IGetTessTriStripEdgeSize Method \(IFace2\)](#)

Gets the size of the array returned by [IFace2::GetTessTriStripEdges](#) and [IFace2::IGetTessTriStripEdges](#).

.NET Syntax

Visual Basic (Declaration)

Function IGetTessTriStripEdgeSize() As System.Integer

Visual Basic (Usage)

```
Dim instance As IFace2
Dim value As System.Integer

value = instance.IGetTessTriStripEdgeSize()
```

C#

System.int IGetTessTriStripEdgeSize()

C++/CLI

System.int IGetTessTriStripEdgeSize();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Size of the array

Visual Basic for Applications (VBA) Syntax

See [Face2::IGetTessTriStripEdgeSize](#).

Remarks

The return value from this method is the number of longs returned from [IFace2::GetTessTriStripEdges](#) and [IFace2::IGetTessTriStripEdges](#), which is (1 + NumStrips + EdgeCount).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetTessNormals Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripNormals Method](#)
[IFace2::IGetTessTriStrips Method](#)
[IFace2::GetTessNormals Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripNormals Method](#)
[IFace2::GetTessTriStrips Method](#)
[IFace2::GetTessTriStripSize Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetTessTriStripNormals Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetTessTriStripNormals Method (IFace2)

Gets the normal vector for each of the triangles that make up the shaded picture tessellation for this face.

.NET Syntax

Visual Basic (Declaration)

Function IGetTessTriStripNormals() As System.Single

Visual Basic (Usage)

```
Dim instance As IFace2
Dim value As System.Single

value = instance.IGetTessTriStripNormals()
```

C#

```
System.float IGetTessTriStripNorms()
```

C++/CLI

```
System.float IGetTessTriStripNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of floats (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The format of the returned data is:

DWORD *NumStrips*

DWORD [*VertexPerStrip*]

Float [*Normals*]

where:

NumStrips = Number of strips on a particular face

VertexPerStrip = Array containing the number of vertex points on each face strip

Normals = Array of X,Y,Z normal components for each vertex on each face strip

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTessNorms Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStrips Method](#)
[IFace2::GetTessTriStripSize Method](#)
[IFace2::IGetTessNorms Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStrips Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetTessTriStrips Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetTessTriStrips Method (IFace2)

Gets the vertices that make up the shaded picture tessellation for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriStrips( _  
    ByVal NoConversion As System.Boolean _  
) As System.Single
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim NoConversion As System.Boolean  
Dim value As System.Single  
  
value = instance.IGetTessTriStrips(NoConversion)
```

C#

```
System.float IGetTessTriStrips(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.float IGetTessTriStrips(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True prohibits conversion to user units from system units, false does not

Return Value

- in-process, unmanaged C++: Pointer to an array of floats (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

These triangles are intended for graphics display purposes and do not represent a tessellation that can be used, for example, by a machining application. If you need the level of accuracy associated with a machining product, traverse the body faces and extract the topology and geometry data to create your own faceting.

This method returns an array of float values in the form of:

[*NumStrips*,

VertexPerStrip1, *VertexPerStrip2*,..., *VertexPerStripN*,

VertexPoints1[*VertexPerStrip1* elements]*3,

VertexPoints2[*VertexPerStrip2* elements]*3,...,

VertexPointsN[*VertexPerStripN* elements]*3]

where:

NumStrips = number of triangle strips on a particular face.

VertexPerStripN = number of vertex points for the *N*th triangle strip.

VertexPointsN = a sub-array of vertices consisting of three float values representing the x,y,z coordinate of the vertex; the sub-array's length is defined by the *VertexPerStripN* element in the array.

Because the values *NumStrips* and the elements of *VertexPerStrip* are long values packed into single values, you must [unpack](#) them before using them.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetTessNorms Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStripNorms Method](#)
[IFace2::GetTessNorms Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStrips Method](#)
[IFace2::GetTessTriStripSize Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTrackingIDs Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetTrackingIDs Method (IFace2)

Gets the [tracking IDs assigned to this face](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetTrackingIDs( _
    ByVal TrackingCookie As System.Integer, _
    ByVal Count As System.Integer, _
    ByRef TrackingIDs As System.Integer _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2
Dim TrackingCookie As System.Integer
Dim Count As System.Integer
Dim TrackingIDs As System.Integer
Dim value As System.Integer

value = instance.IGetTrackingIDs(TrackingCookie, Count, TrackingIDs)
```

C#

```
System.int IGetTrackingIDs(
    System.int TrackingCookie,
    System.int Count,
    out System.int TrackingIDs
)
```

C++/CLI

```
System.int IGetTrackingIDs(
    & System.int TrackingCookie,
    & System.int Count,
    & [Out] System.int TrackingIDs
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie

Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

Count

Number of tracking IDs on this face

TrackingIDs

- in-process, unmanaged C++: Pointer to an array of tracking IDs on this face

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

Status as defined by [swTrackingIDError_e](#)

Remarks

Before calling this method, call [IFace2::GetTrackingIDsCount](#) to get Count.

See [Tracking IDs](#) for more information.

See Also

[IFace2 Interface](#)

[IFace2 Members](#)

[IFace2::GetTrackingIDs Method](#)

[IFace2::RemoveTrackingID Method](#)

[IFace2::SetTrackingID Method](#)

[ModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTrimCurveSize Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : [IGetTrimCurveSize Method \(IFace2\)](#)

Obsolete. Superseded by [IFace2::IGetTrimCurveSize2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTrimCurveSize( _  
    ByVal WantCubic As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim WantCubic As System.Boolean  
Dim value As System.Integer  
  
value = instance.IGetTrimCurveSize(WantCubic)
```

C#

```
System.int IGetTrimCurveSize(  
    System.bool WantCubic  
)
```

C++/CLI

```
System.int IGetTrimCurveSize(  
& System.bool WantCubic  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubic

Visual Basic for Applications (VBA) Syntax

See [Face2::IGetTrimCurveSize](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetTrimCurveSize2 Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetTrimCurveSize2 Method (IFace2)

Gets the size of the array required for [IFace2::GetTrimCurves2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTrimCurveSize2(  
    ByVal WantCubic As System.Integer, _  
    ByVal WantNRational As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim WantCubic As System.Integer  
Dim WantNRational As System.Integer  
Dim value As System.Integer  
  
value = instance.IGetTrimCurveSize2(WantCubic, WantNRational)
```

C#

```
System.int IGetTrimCurveSize2(  
    System.int WantCubic,  
    System.int WantNRational  
)
```

C++/CLI

```
System.int IGetTrimCurveSize2(
& System.int WantCubic,
& System.int WantNRational
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubic

True if the trim curves are to be cubic, false if not

WantNRational

True if the trim curves are to be non-rational, false if not

Return Value

Size of the array required for IFace2::GetTrimCurves2

Visual Basic for Applications (VBA) Syntax

See [Face2::IGetTrimCurveSize2](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTrimCurveTopology Method](#)
[IFace2::GetTrimCurveTopologyCount Method](#)
[IFace2::GetTrimCurveTopologyTypes Method](#)
[IFace2::IGetTrimCurveTopology Method](#)
[IFace2::IGetTrimCurveTopologyTypes Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTrimCurveTopology Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetTrimCurveTopology Method (IFace2)

Gets the trim curve topology for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTrimCurveTopology() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2
Dim value As System.Object

value = instance.IGetTrimCurveTopology()
```

C#

```
System.object IGetTrimCurveTopology()
```

C++/CLI

```
System.Object^ IGetTrimCurveTopology();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of coedges, vertices, and NULLs (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This method returns a list of coedges, vertices, and NULLs. You must call [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) before calling this method.

This method returns a list of proper edges corresponding to the list of SP curves from IFace2::GetTrimCurves2 or IFace2::IGetTrimCurves2. Otherwise, this method prompts the user for the list of edges that the software thinks are on the face. These two lists differ in order and content.

Use [IFace2::GetTrimCurveTopologyTypes](#) or [IFace2::IGetTrimCurveTopologyTypes](#) to get an array of types corresponding to the returned objects.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetTrimCurveSize2 Method](#)
[IFace2::GetTrimCurveTopologyCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTrimCurveTopologyTypes Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetTrimCurveTopologyTypes Method (IFace2)

Gets the trim curve topology type array for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTrimCurveTopologyTypes() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Integer  
  
value = instance.IGetTrimCurveTopologyTypes()
```

C#

```
System.int IGetTrimCurveTopologyTypes()
```

C++/CLI

```
System.int IGetTrimCurveTopologyTypes();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of longs indicating the topology types as defined in [swSelectType_e](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

Use [IFace2::GetTrimCurveTopology](#) or [IFace2::IGetTrimCurveTopology](#) to get the trim curve topology for this face.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTrimCurves2 Method](#)
[IFace2::GetTrimCurveTopologyCount Method](#)
[IFace2::IGetTrimCurveSize2 Method](#)
[IFace2::GetTrimCurveTopology Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetUVBounds Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IGetUVBounds Method (IFace2)

Gets the values that describe the U, V bounds of this face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetUVBounds() As System.Double
```

Visual Basic (Usage)

```
Dim instance As IFace2  
Dim value As System.Double  
  
value = instance.IGetUVBounds()
```

C#

```
System.double IGetUVBounds()
```

C++/CLI

```
System.double IGetUVBounds();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return values bound an area of the surface in which the face is defined. You can determine the surface U, V bounds using [ISurface::Parameterization](#) or [ISurface::IParameterization](#).

The returned data is an array of 4 doubles arranged in the following order:

retval[0] - Minimum U parameter of this face
retval[1] - Maximum U parameter of this face
retval[2] - Minimum V parameter of this face
retval[3] - Maximum V parameter of this face

The minimum parameters are always less than the maximum parameters, and the range (for example, retval[1] - retval[0] and retval[3]-retval[2]) is always less than or equal to the U and V range of the underlying surface.

For surfaces with periodic parameters, the face parameter box can never be larger than the period of the corresponding surface parameter.

uRange[0] <= retval[0] < uRange[1]
vRange[0] <= retval[2] < vRange[1]

where uRange and vRange describe the UV range of the surface

Therefore, a face that straddles the boundary of a periodic parameter has an upper parameter value for the face that is greater than the upper parameter range of the surface.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IBody2::GetProcessedBody2 Method](#)
[IFace2::GetUVBounds Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IHighlight Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IHighlight Method (IFace2)

Adds highlighting to or removes highlighting from a face.

.NET Syntax

Visual Basic (Declaration)

```
Sub IHighlight( _  
    ByVal State As System.Boolean _  
)
```

```
Dim instance As IFace2
Dim State As System.Boolean

instance.IHighlight(State)
```

```
void IHighlight(
    System.bool State
)
```

```
void IHighlight(
    & System.bool State
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

State

True adds highlighting, false removes highlighting

Visual Basic for Applications (VBA) Syntax

See [Face2.IHighlight](#).

Remarks

Highlighting remains in effect until the model is rebuilt or redrawn.

This method is not supported for faces obtained from reference surface bodies.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2.Highlight Method](#)
[Edge.Highlight Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ImprintCurve Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : ImprintCurve Method (IFace2)

Imprints a curve on the selected face.

.NET Syntax

```
Sub IImprintCurve( _
    ByVal Curve As Curve, _
    ByVal NewEdgeCount As System.Integer, _
    ByRef NewEdges As Edge, _
    ByVal NewFaceCount As System.Integer, _
    ByRef NewFaces As Face2 _
)
```

```
Dim instance As IFace2
Dim Curve As Curve
Dim NewEdgeCount As System.Integer
Dim NewEdges As Edge
Dim NewFaceCount As System.Integer
```

Dim NewFaces As Face2

instance.IImprintCurve(Curve, NewEdgeCount, NewEdges, NewFaceCount, NewFaces)

C#

```
void IImprintCurve(  
    Curve Curve,  
    System.int NewEdgeCount,  
    out Edge NewEdges,  
    System.int NewFaceCount,  
    out Face2 NewFaces  
)
```

C++/CLI

```
void IImprintCurve(  
& Curve^ Curve,  
& System.int NewEdgeCount,  
& [Out] Edge^ NewEdges,  
& System.int NewFaceCount,  
& [Out] Face2^ NewFaces  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Curve

Pointer to the [curve](#) to imprint on the face

NewEdgeCount

Number of new edges to create

NewEdges

Array of new [edges](#) created by the imprinted curve of size NewEdgeCount

NewFaceCount

Number of new faces to create

NewFaces

Array of new [faces](#) created by the imprinted curve of size NewFaceCount

Visual Basic for Applications (VBA) Syntax

See [Face2::IImprintCurve](#).

Remarks

Before calling this method, call [IFace2::ImprintCurveCount](#) to get the size of the arrays.

The specified curve must lie on the face.

To imprint a curve on a new face of a temporary body, create a copy of the original curve and imprint the copy of the curve on the new face.

See Also

[IFace2 Interface](#)

[IFace2 Members](#)

[IFace2::ImprintCurve Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

IIsCoincident Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IIsCoincident Method (IFace2)

Gets whether this face and the specified face, which is located on a different body, are coincident.

.NET Syntax

Visual Basic (Declaration)

```
Function IIsCoincident( _
    ByVal FaceIn As Face2, _
    ByVal Tolerance As System.Double _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2
Dim FaceIn As Face2
Dim Tolerance As System.Double
Dim value As System.Integer

value = instance.IIsCoincident(FaceIn, Tolerance)
```

C#

```
System.int IIsCoincident(
    Face2 FaceIn,
    System.double Tolerance
)
```

C++/CLI

```
System.int IIsCoincident(
    & Face2^ FaceIn,
    & System.double Tolerance
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceIn
[Face](#) located on a different body
Tolerance
Tolerance

Return Value

Result as defined in [swFaceCoincidentResult](#) e (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2.IIsCoincident](#).

Remarks

For two faces to be considered coincident, they must have similar corresponding loops and all points on one face must be within the specified tolerance of the other face, and vice versa.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2.IIsCoincident Method](#)

Availability

SOLIDWORKS 2007 SP3, Revision Number 15.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

IIsSame Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IIsSame Method (IFace2)

Gets whether the two faces are the same.

.NET Syntax

Visual Basic (Declaration)

```
Function IIsSame( _
    ByVal FaceIn As Face2 _
) As System.Boolean
```

```
Dim instance As IFace2
Dim FaceIn As Face2
Dim value As System.Boolean

value = instance.IIsSame(FaceIn)
```

```
System.bool IIsSame(
    Face2 FaceIn
)
```

```
System.bool IIsSame(
    & Face2^ FaceIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceIn

Pointer to the [face](#) to which to compare this face

Return Value

True if the two faces are the same, false if they are different

Visual Basic for Applications (VBA) Syntax

See [Face2.IIsSame](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2.IIsSame Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MaterialPropertyValues Property (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : IMaterialPropertyValues Property (IFace2)

Gets or sets the material properties for the selected face in the active configuration.

.NET Syntax

```
Property IMaterialPropertyValues As System.Double
```

```
Dim instance As IFace2
Dim value As System.Double

instance.IMaterialPropertyValues = value

value = instance.IMaterialPropertyValues
```

```
System.double IMaterialPropertyValues {get; set;}
```

```
property System.double IMaterialPropertyValues {
    System.double get();
```

```
void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles describing the material values on this face (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::IMaterialPropertyValues](#).

Remarks

The material values include the face color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency and emission. All values can be from 0 to 1.

This method returns a NULL VARIANT (or an S_FALSE HRESULT for COM implementations) if this face has not been explicitly modified from the material property values of the body. If you create a body and change the body color to red, then [IFace2::IMaterialPropertyValues2](#) and [IFace2::GetMaterialPropertyValues2](#) return a NULL array because you did not specifically change the values of the face.

The format of the parameters or return values is an array of doubles as follows:

[*R*, *G*, *B*, *Ambient*, *Diffuse*, *Specular*, *Shininess*, *Transparency*, *Emission*]

1. Specify *R*, *G*, and *B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse*, *Specular*, *Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient*, *Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after setting this property.

To reset the face to use the default part material properties, use [IModelDoc2::SelectedFaceProperties](#).

This method does not support faces obtained from reference surface bodies.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::MaterialPropertyValues Property](#)
[IFace2::IRemoveMaterialProperty2 Method](#)
[IFace2::ISetMaterialPropertyValues2 Method](#)
[IFace2::RemoveMaterialProperty2 Method](#)
[IFace2::SetMaterialPropertyValues2 Method](#)
[IBody2::IMaterialPropertyValues2 Property](#)
[IBody2::MaterialPropertyValues2 Property](#)
[IComponent2::IMaterialPropertyValues Property](#)
[IComponent2::MaterialPropertyValues Property](#)
[IModelDoc2::IMaterialPropertyValues Property](#)
[IModelDoc2::MaterialPropertyValues Property](#)
[IPartDoc::IMaterialPropertyValues Property](#)
[IPartDoc::MaterialPropertyValues Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ImprintCurve Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : ImprintCurve Method (IFace2)

Imprints a curve on the selected face.

.NET Syntax

Visual Basic (Declaration)

```
Sub ImprintCurve(  
    ByVal Curve As System.Object, _  
    ByRef NewEdges As System.Object, _  
    ByRef NewFaces As System.Object _  
)
```

Visual Basic (Usage)

```
Dim instance As IFace2  
Dim Curve As System.Object  
Dim NewEdges As System.Object  
Dim NewFaces As System.Object  
  
instance.ImprintCurve(Curve, NewEdges, NewFaces)
```

C#



```
void ImprintCurve(  
    System.Object Curve,  
    out System.Object NewEdges,  
    out System.Object NewFaces  
)
```

C++/CLI

```
void ImprintCurve(  
& System.Object^ Curve,  
& [Out] System.Object^ NewEdges,  
& [Out] System.Object^ NewFaces  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Curve

[Curve](#) to imprint on the face

NewEdges

Array of new [edges](#) created by the imprinted curve

NewFaces

Array of new [faces](#) created by the imprinted curve

Visual Basic for Applications (VBA) Syntax

See [Face2::ImprintCurve](#).

Remarks

The specified curve must lie on the face.

To imprint a curve on a new face of a temporary body, create a copy of the original curve and imprint the copy of the curve on the new face.

See Also

[IFace2 Interface](#)

[IFace2 Members](#)

[IFace2::ImprintCurve Method](#)

[IFace2::ImprintCurveCount Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2